

Nickname:

1.9 Classwork: Bivariate data and linear regression

Use your GDC. Give answers exactly, or correct to three significant figures. Write down the values you put into the calculator.

1. A gelateria in Rome recorded the maximum temperature, x °C, and the number of gelati sold, y , on eight days in June.

Temperature, x (°C)	18	20	21	23	25	28	30	32
Gelati sold, y	150	158	181	185	214	226	262	258

- (a) Find Pearson's product-moment correlation coefficient, r .
- (b) Describe the correlation between x and y (direction: positive or negative; strength: strong, moderate, weak or none).
- (c) Write down the equation of the regression line of y on x , in the form $y = ax + b$.
- (d) Interpret the meaning of the value of a in this context.
- (e) Use your regression line to estimate the number of gelati sold on a day when the maximum temperature is 26 °C.
- (f) Explain why the regression line should not be used to estimate the number of gelati sold on a day in January when the maximum temperature is 10 °C.

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2. A student asked eight classmates how many hours a day they spend on their phone, x , and how many hours they sleep a night, y .

Phone time, x (hours)	1.5	2	2.5	3	4	4.5	5.5	6
Sleep, y (hours)	8.2	8.7	7.6	8.1	7.0	7.8	6.6	7.1

- Find the equation of the regression line of y on x , in the form $y = ax + b$.
- Interpret the meaning of the value of b in this context.
- Find the coordinates of the mean point, $(\bar{x}, \bar{y})$.
- Find the equation of the regression line of x on y .
- Another student, Lucia, sleeps 7.0 hours a night. Use the appropriate regression line to estimate her daily phone time.